Experiment/ Case Study No.: 3

1. Experiment Title: Implement JavaScript control structures and form validation by developing a grading system based on user-entered marks.

2. Software/Tools Required:

- Visual Studio Code (VS Code) / Notepad++
- Google Chrome / Mozilla Firefox
- JavaScript (ES6)
- HTML5
- Windows/Linux Operating System

3. Experiment Program Code:

1. practical3.html:

```
<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Student Performance Analytics Dashboard</title>

<link rel="preconnect" href="https://fonts.googleapis.com">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;7
00&display=swap" rel="stylesheet">

<link rel="stylesheet" href="practical3.css">

</head>

<body>

<div class="dashboard">

  <!-- Sidebar -->
```

```
<div class="sidebar">
  <h2>🎓 Dashboard</h2>
  <ul>
    <li>🏠 Home</li>
    <li>👤 Student</li>
    <li>📊 Analytics</li>
    <li>🏆 Report</li>
    <li>⚙ Settings</li>
  </ul>
</div>
<!-- Main Content -->
<div class="main">
  <div class="topbar">
    <h1>Student Performance Analytics</h1>
    <button onclick="toggleTheme()">🌙 Dark Mode</button>
  </div>
  <!-- Student Details -->
  <div class="student-form">
    <h2>Student Information</h2>
    <div class="grid">
      <input type="text" id="name" placeholder="Student Name">
      <input type="text" id="roll" placeholder="Roll Number">
      <input type="text" id="class" placeholder="Class">
      <select id="semester">
```

```
      <option value="">Select Semester</option>

      <option>Semester I</option>
      <option>Semester II</option>
      <option>Semester III</option>
      <option>Semester IV</option>
      <option>Semester V</option>
      <option>Semester VI</option>

    </select>

</div>

<h2>Enter Subject Marks</h2>

<table>

    <tr>

      <th>Subject</th>

      <th>Marks</th>

    </tr>

    <tr>

      <td>English</td>

      <td><input type="number" id="english" min="0" max="100"></td>

    </tr>

    <tr>

      <td>Mathematics</td>

      <td><input type="number" id="maths" min="0" max="100"></td>

    </tr>

    <tr>

      <td>Science</td>

      <td><input type="number" id="science" min="0" max="100"></td>

    </tr>

    <tr>
```

```
            <td>Computer</td>
            <td><input type="number" id="computer" min="0" max="100"></td>
        </tr>
        <tr>
            <td>Social Science</td>
            <td><input type="number" id="social" min="0" max="100"></td>
        </tr>
    </table>
    <div class="buttons">
        <button onclick="analyzePerformance()">📊 Analyze</button>
        <button onclick="resetForm()">🔄 Reset</button>
        <button onclick="printReport()">🖨 Print</button>
    </div>
</div>
<!-- Dashboard Cards -->
<div class="cards">
    <div class="card">
        <h3>Total Marks</h3>
        <h1 id="total">0</h1>
    </div>
    <div class="card">
        <h3>Percentage</h3>
        <h1 id="percentage">0%</h1>
    </div>
```

```
            <div class="card">
              <h3>Grade</h3>
              <h1 id="grade">-</h1>
            </div>
            <div class="card">
              <h3>Status</h3>
              <h1 id="status">-</h1>
            </div>
          </div>
          <!-- Analytics -->
          <div class="analytics">
            <h2>Performance Analytics</h2>
            <p>Highest Marks : <span id="highest">-</span></p>
            <p>Lowest Marks : <span id="lowest">-</span></p>
            <p>Average : <span id="average">-</span></p>
            <p>Scholarship : <span id="scholarship">-</span></p>
            <p>Teacher Remark : <span id="remarks">-</span></p>
            <div class="progress">
              <div id="progressBar">0%</div>
            </div>
          </div>
        </div>
      </div>
      <script src="practical3.js"></script>
      </body>
</html>
```

2. practical3.css

```
/* ========================================
  Student Performance Analytics Dashboard
  style.css
======================================== */

*{
   margin:0;
   padding:0;
   box-sizing:border-box;
   font-family:'Poppins',sans-serif;
}

body{
   background:#eef2f7;
}

/* Dashboard Layout */

.dashboard{
   display:flex;
   min-height:100vh;
}

/* Sidebar */

.sidebar{
   width:240px;
   background:#1f2937;
   color:white;
   padding:25px;
}

.sidebar h2{
   text-align:center;
   margin-bottom:30px;
}

.sidebar ul{
   list-style:none;
}

.sidebar ul li{
   padding:15px;
   margin:10px 0;
   border-radius:10px;
   cursor:pointer;
   transition:.3s;
}
```

```
.sidebar ul li:hover{
   background:#374151;
}

/* Main Section */

.main{
   flex:1;
   padding:30px;
}

/* Top Bar */

.topbar{
   display:flex;
   justify-content:space-between;
   align-items:center;
   margin-bottom:25px;
}

.topbar h1{
   color:#1f2937;
}

.topbar button{
   background:#2563eb;
   color:white;
   border:none;
   padding:10px 18px;
   border-radius:8px;
   cursor:pointer;
   font-weight:bold;
}

.topbar button:hover{
   background:#1d4ed8;
}

/* Student Form */

.student-form{

   background:white;

   padding:25px;

   border-radius:15px;

   box-shadow:0 5px 15px rgba(0,0,0,.1);
```

```
    margin-bottom:30px;
}
.student-form h2{
    margin-bottom:15px;
    color:#2563eb;
}
.grid{
    display:grid;
    grid-template-columns:repeat(2,1fr);
    gap:15px;
    margin-bottom:20px;
}
.grid input,
.grid select{
    padding:12px;
    border:1px solid #ccc;
    border-radius:8px;
    outline:none;
}
table{
    width:100%;
    border-collapse:collapse;
    margin-top:15px;
}
table th{
```

```
    background:#2563eb;
    color:white;
    padding:12px;
}
table td{
    padding:12px;
    border:1px solid #ddd;
}
table input{
    width:100%;
    padding:8px;
    border:none;
    outline:none;
}
/* Buttons */
.buttons{
    margin-top:20px;
    display:flex;
    gap:15px;
}
.buttons button{
    flex:1;
    padding:12px;
    border:none;
    border-radius:8px;
```

```
    font-weight:bold;
    cursor:pointer;
    transition:.3s;
}
.buttons button:nth-child(1){
    background:#16a34a;
    color:white;
}
.buttons button:nth-child(2){
    background:#f59e0b;
    color:white;
}
.buttons button:nth-child(3){
    background:#dc2626;
    color:white;
}
.buttons button:hover{
    transform:translateY(-2px);
}
/* Dashboard Cards */
.cards{
    display:grid;
    grid-template-columns:repeat(4,1fr);
    gap:20px;
    margin-bottom:25px;
```

```
}
.card{
    background:white;
    border-radius:15px;
    padding:25px;
    text-align:center;
    box-shadow:0 5px 15px rgba(0,0,0,.1);
    transition:.3s;
}
.card:hover{
    transform:translateY(-5px);
}
.card h3{
    color:#666;
    margin-bottom:10px;
}
.card h1{
    color:#2563eb;
    font-size:35px;
}
/* Analytics Section */
.analytics{
    background:white;
    border-radius:15px;
    padding:25px;
```

```
    box-shadow:0 5px 15px rgba(0,0,0,.1);
}
.analytics h2{
    margin-bottom:20px;
    color:#2563eb;
}
.analytics p{
    margin:12px 0;
    font-size:17px;
}
/* Progress Bar */
.progress{
    width:100%;
    height:28px;
    background:#ddd;
    border-radius:20px;
    overflow:hidden;
    margin-top:20px;
}
#progressBar{
    width:0%;
    height:100%;
    background:linear-gradient(90deg,#16a34a,#22c55e);
    color:white;
    text-align:center;
```

```
    line-height:28px;

    font-weight:bold;

    transition:1s;

}

/* Dark Mode */

.dark{

    background:#111827;

}

.dark .student-form,
.dark .analytics,
.dark .card{

    background:#1f2937;

    color:white;

}

.dark .topbar h1{

    color:white;

}

.dark table td{

    background:#374151;

    color:white;

}

.dark table input{

    background:#374151;

    color:white;

}

.dark .grid input,
.dark .grid select{
```

```
    background:#374151;
    color:white;
    border:none;
}
/* Responsive */
@media(max-width:900px){
.dashboard{
flex-direction:column;
}
.sidebar{
width:100%;
}
.cards{
grid-template-columns:repeat(2,1fr);
}
.grid{
grid-template-columns:1fr;
}
}
@media(max-width:600px){
.cards{
grid-template-columns:1fr;
}
.buttons{
flex-direction:column;
```

```
}

}
```

3. practical3.js

```
// ======================================
// Student Performance Analytics Dashboard
// script.js
// ======================================

// var
var collegeName = "Student Performance Dashboard";

// const
const PASS_MARK = 35;
const TOTAL_MARKS = 500;

// Analyze Performance

function analyzePerformance() {

  // Student Details

  let name = document.getElementById("name").value.trim();
  let roll = document.getElementById("roll").value.trim();
  let studentClass = document.getElementById("class").value.trim();
  let semester = document.getElementById("semester").value;

  // Marks

  let english = Number(document.getElementById("english").value);
  let maths = Number(document.getElementById("maths").value);
  let science = Number(document.getElementById("science").value);
  let computer = Number(document.getElementById("computer").value);
  let social = Number(document.getElementById("social").value);

  // Form Validation

  if (
    name === "" ||
    roll === "" ||
    studentClass === "" ||
    semester === ""
  ) {
    alert("Please fill all student details.");
    return;
  }

  let marks = [english, maths, science, computer, social];
```

```
for (let i = 0; i < marks.length; i++) {
    if (isNaN(marks[i]) || marks[i] < 0 || marks[i] > 100) {
        alert("Marks should be between 0 and 100.");
        return;
    }
}
// Calculations
let total = marks.reduce((sum, mark) => sum + mark, 0);
let percentage = total / 5;
let highest = Math.max(...marks);
let lowest = Math.min(...marks);
let average = (total / 5).toFixed(2);
// Pass / Fail
let status = "PASS";
for (let i = 0; i < marks.length; i++) {
    if (marks[i] < PASS_MARK) {
        status = "FAIL";
        break;
    }
}
// Grade
let grade;
if (percentage >= 90)
    grade = "A+";
else if (percentage >= 80)
```

```
    grade = "A";
else if (percentage >= 70)
    grade = "B+";
else if (percentage >= 60)
    grade = "B";
else if (percentage >= 50)
    grade = "C";
else if (percentage >= 35)
    grade = "D";
else
    grade = "F";
// Remarks using switch
let remarks;
switch (grade) {
    case "A+":
        remarks = "Outstanding Performance 🌟";
        break;
    case "A":
        remarks = "Excellent Work 🎉";
        break;
    case "B+":
        remarks = "Very Good 👍";
        break;
    case "B":
```

```
        remarks = "Good Performance";
        break;
    case "C":
        remarks = "Average. Keep Improving.";
        break;
    case "D":
        remarks = "Needs More Practice.";
        break;
    default:
        remarks = "Failed. Work Hard.";
}

// Scholarship
let scholarship;
if (percentage >= 85)
    scholarship = "Eligible 🏆 ";
else
    scholarship = "Not Eligible";

// Object
let student = {
    studentName: name,
    rollNumber: roll,
    className: studentClass,
    semesterName: semester,
    totalMarks: total,
    percentage: percentage,
    highestMarks: highest,
    lowestMarks: lowest,
    averageMarks: average,
    studentGrade: grade,
    studentStatus: status,
    studentRemarks: remarks,
```

```
    studentScholarship: scholarship
  };
  // Object Destructuring
  const {
    totalMarks,
    highestMarks,
    lowestMarks,
    averageMarks,
    studentGrade,
    studentStatus,
    studentRemarks,
    studentScholarship
  } = student;
  // Display Dashboard
  document.getElementById("total").innerHTML = totalMarks;
  document.getElementById("percentage").innerHTML = percentage.toFixed(2) +
"%";
  document.getElementById("grade").innerHTML = studentGrade;
  document.getElementById("status").innerHTML = studentStatus;
  document.getElementById("highest").innerHTML = highestMarks;
  document.getElementById("lowest").innerHTML = lowestMarks;
  document.getElementById("average").innerHTML = averageMarks;
  document.getElementById("scholarship").innerHTML = studentScholarship;
  document.getElementById("remarks").innerHTML = studentRemarks;
  // Progress Bar
  let progress = document.getElementById("progressBar");
```

```
    progress.style.width = percentage + "%";

    progress.innerHTML = percentage.toFixed(0) + "%";

    // Status Color

    if (studentStatus === "PASS") {

        document.getElementById("status").style.color = "green";

    } else {

        document.getElementById("status").style.color = "red";

    }

    // Template Literal

    console.log(`Student ${name} scored ${percentage.toFixed(2)}%`);

    console.table(student);

}

// Reset

function resetForm() {

    document.getElementById("name").value = "";
    document.getElementById("roll").value = "";
    document.getElementById("class").value = "";
    document.getElementById("semester").selectedIndex = 0;

    document.getElementById("english").value = "";
    document.getElementById("maths").value = "";
    document.getElementById("science").value = "";
    document.getElementById("computer").value = "";
    document.getElementById("social").value = "";

    document.getElementById("total").innerHTML = "0";
    document.getElementById("percentage").innerHTML = "0%";
    document.getElementById("grade").innerHTML = "-";
    document.getElementById("status").innerHTML = "-";
    document.getElementById("highest").innerHTML = "-";
    document.getElementById("lowest").innerHTML = "-";
    document.getElementById("average").innerHTML = "-";
    document.getElementById("scholarship").innerHTML = "-";
    document.getElementById("remarks").innerHTML = "-";
```

```
    let progress = document.getElementById("progressBar");
    progress.style.width = "0%";
    progress.innerHTML = "0%";

}

// Print Dashboard

function printReport() {

    window.print();

}

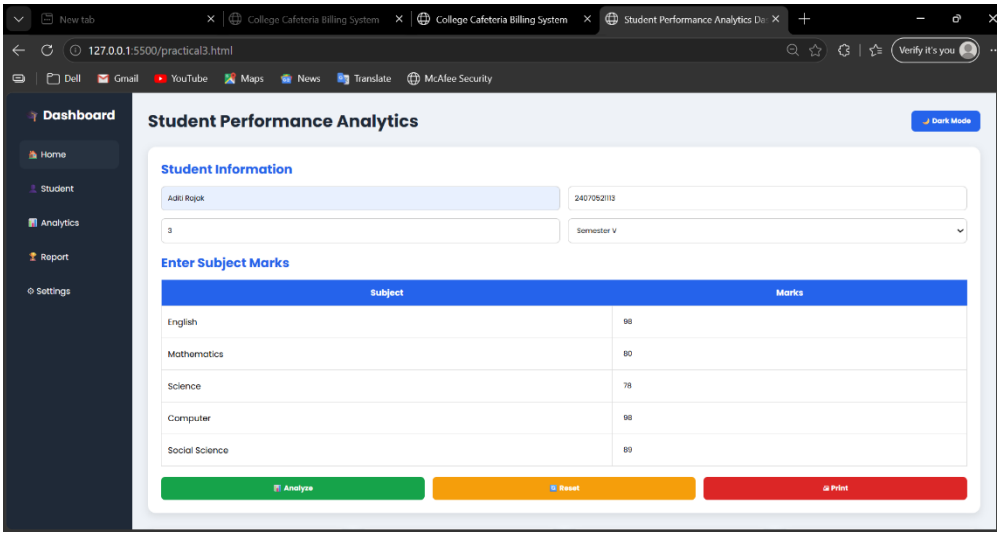
// Dark Mode

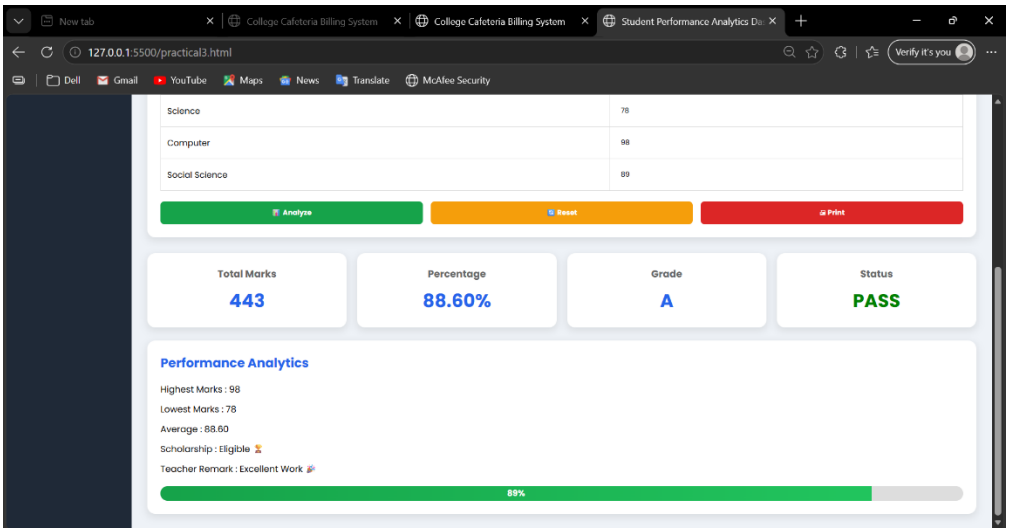
function toggleTheme() {

    document.body.classList.toggle("dark");

}
```

4. Output:





[Student Performance Analytics Dashboard]

5. Case Study Title: **Employee Performance Evaluation System**

6. Case Study Program Code:

1. caseStudy3.html:

```
<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Employee Performance Evaluation System</title>

    <link rel="preconnect" href="https://fonts.googleapis.com">

    <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap" rel="stylesheet">

    <link rel="stylesheet" href="caseStudy3.css">

</head>

<body>

<div class="container">
```

```
<div class="header">
  <h1>💼 Employee Performance Evaluation</h1>
  <p>HR Performance Management Dashboard</p>
</div>
<button class="theme-btn" onclick="toggleTheme()">🌙 Dark Mode</button>
<div class="form-section">
  <h2>Employee Information</h2>
  <div class="grid">
    <input type="text" id="empName" placeholder="Employee Name">
    <input type="text" id="empId" placeholder="Employee ID">
    <input type="text" id="department" placeholder="Department">
    <select id="designation">
      <option value="">Select Designation</option>
      <option>Software Engineer</option>
      <option>Team Leader</option>
      <option>HR Executive</option>
      <option>Project Manager</option>
      <option>Data Analyst</option>
    </select>
  </div>
  <h2>Performance Scores (0-100)</h2>
  <table>
    <tr>
      <th>Criteria</th>
      <th>Score</th>
```

```
        </tr>
        <tr>
          <td>Technical Skills</td>
          <td><input type="number" id="technical" min="0" max="100"></td>
        </tr>
        <tr>
          <td>Communication</td>
          <td><input type="number" id="communication" min="0"
max="100"></td>
        </tr>
        <tr>
          <td>Teamwork</td>
          <td><input type="number" id="teamwork" min="0" max="100"></td>
        </tr>
        <tr>
          <td>Productivity</td>
          <td><input type="number" id="productivity" min="0" max="100"></td>
        </tr>
        <tr>
          <td>Attendance</td>
          <td><input type="number" id="attendance" min="0" max="100"></td>
        </tr>
      </table>
      <div class="buttons">
        <button onclick="evaluateEmployee()">📊 Evaluate</button>
```

```
      <button onclick="resetForm()">🔄 Reset</button>
      <button onclick="printReport()">🖨 Print</button>
    </div>
  </div>
  <div class="report">
    <h2>📋 Performance Report</h2>
    <div class="cards">
      <div class="card">
        <h3>Total Score</h3>
        <h1 id="total">0</h1>
      </div>
      <div class="card">
        <h3>Average</h3>
        <h1 id="average">0</h1>
      </div>
      <div class="card">
        <h3>Grade</h3>
        <h1 id="grade">-</h1>
      </div>
      <div class="card">
        <h3>Status</h3>
        <h1 id="status">-</h1>
      </div>
    </div>
    <table class="summary">
```

```
    <tr>
      <td>Highest Score</td>
      <td id="highest">-</td>
    </tr>
    <tr>
      <td>Lowest Score</td>
      <td id="lowest">-</td>
    </tr>
    <tr>
      <td>Promotion Eligibility</td>
      <td id="promotion">-</td>
    </tr>
    <tr>
      <td>Bonus Eligibility</td>
      <td id="bonus">-</td>
    </tr>
    <tr>
      <td>HR Remarks</td>
      <td id="remarks">-</td>
    </tr>
  </table>
  <h3>Performance Meter</h3>
  <div class="progress">
    <div id="progressBar">0%</div>
  </div>
```

```
    </div>

</div>

<script src="caseStudy3.js"></script>

</body>

</html>
```

2. caseStudy2.css:

```
/* ========================================
   Employee Performance Evaluation System
   style.css
======================================== */

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:'Poppins',sans-serif;
}

body{
    background:#eef2f7;
}

/* Container */

.container{
    width:90%;
    max-width:1200px;
    margin:30px auto;
}

/* Header */

.header{
    text-align:center;
    margin-bottom:20px;
}

.header h1{
    color:#1e3a8a;
    font-size:34px;
}

.header p{
    color:#555;
```

```
   margin-top:8px;
}

/* Dark Mode Button */

.theme-btn{
   float:right;
   margin-bottom:20px;
   background:#2563eb;
   color:white;
   border:none;
   padding:10px 18px;
   border-radius:8px;
   cursor:pointer;
   transition:.3s;
}

.theme-btn:hover{
   background:#1d4ed8;
}

/* Form Section */

.form-section{

   clear:both;

   background:white;

   padding:25px;

   border-radius:15px;

   box-shadow:0 5px 15px rgba(0,0,0,.1);

}

.form-section h2{

   color:#2563eb;

   margin-bottom:15px;

}

.grid{

   display:grid;

   grid-template-columns:repeat(2,1fr);
```

```
    gap:15px;

    margin-bottom:25px;

}

.grid input,
.grid select{

    width:100%;

    padding:12px;

    border:1px solid #ccc;

    border-radius:8px;

    outline:none;

    font-size:15px;

}

.grid input:focus,
.grid select:focus{

    border-color:#2563eb;

}

/* Table */

table{

    width:100%;

    border-collapse:collapse;

    margin-top:15px;

}

table th{

    background:#2563eb;

    color:white;

    padding:12px;
```

```
}
table td{
    padding:12px;
    border:1px solid #ddd;
}
table input{
    width:100%;
    padding:8px;
    border:none;
    outline:none;
}
/* Buttons */
.buttons{
    margin-top:20px;
    display:flex;
    gap:15px;
}
.buttons button{
    flex:1;
    padding:12px;
    border:none;
    border-radius:8px;
    color:white;
    font-size:15px;
    font-weight:bold;
```

```
    cursor:pointer;
    transition:.3s;
}
.buttons button:nth-child(1){
    background:#16a34a;
}
.buttons button:nth-child(2){
    background:#f59e0b;
}
.buttons button:nth-child(3){
    background:#dc2626;
}
.buttons button:hover{
    transform:translateY(-3px);
}
/* Report */
.report{
    margin-top:30px;
    background:white;
    border-radius:15px;
    padding:25px;
    box-shadow:0 5px 15px rgba(0,0,0,.1);
}
.report h2{
    color:#2563eb;
```

```
    margin-bottom:20px;
}
/* Cards */
.cards{
    display:grid;
    grid-template-columns:repeat(4,1fr);
    gap:20px;
    margin-bottom:20px;
}
.card{
    background:#f8fafc;
    border-left:6px solid #2563eb;
    border-radius:10px;
    padding:20px;
    text-align:center;
    transition:.3s;
}
.card:hover{
    transform:translateY(-5px);
}
.card h3{
    color:#555;
}
.card h1{
    color:#2563eb;
```

```
    margin-top:10px;
    font-size:30px;
}
/* Summary */
.summary{
    margin-top:20px;
}
.summary td{
    padding:12px;
    font-size:16px;
}
/* Progress Bar */
.progress{
    width:100%;
    height:30px;
    background:#ddd;
    border-radius:20px;
    overflow:hidden;
    margin-top:15px;
}
#progressBar{
    width:0%;
    height:100%;
    background:linear-gradient(90deg,#16a34a,#22c55e);
    color:white;
```

```
    text-align:center;

    line-height:30px;

    font-weight:bold;

    transition:1s;

}

/* Dark Mode */

.dark{

    background:#111827;

}

.dark .form-section,
.dark .report{

    background:#1f2937;

    color:white;

}

.dark table td{

    background:#374151;

    color:white;

}

.dark input,
.dark select{

    background:#374151;

    color:white;

    border:none;

}

.dark .card{

    background:#374151;
```

```
    color:white;
}
.dark .header h1{
    color:white;
}
.dark .header p{
    color:#ddd;
}
/* Responsive */
@media(max-width:900px){
.cards{
grid-template-columns:repeat(2,1fr);
}
.grid{
grid-template-columns:1fr;
}
}
@media(max-width:600px){
.cards{
grid-template-columns:1fr;
}
.buttons{
flex-direction:column;
}
}
```

3. caseStudy3.js:

```
// ========================================
// Employee Performance Evaluation System
// script.js
// ========================================

// var
var companyName = "ABC Technologies Pvt. Ltd.";

// const
const MAX_SCORE = 500;
const PASS_SCORE = 35;

// Evaluate Employee

function evaluateEmployee() {

   // Employee Details

   let empName = document.getElementById("empName").value.trim();
   let empId = document.getElementById("empId").value.trim();
   let department = document.getElementById("department").value.trim();
   let designation = document.getElementById("designation").value;

   // Performance Scores

   let technical = Number(document.getElementById("technical").value);
   let communication =
Number(document.getElementById("communication").value);
   let teamwork = Number(document.getElementById("teamwork").value);
   let productivity = Number(document.getElementById("productivity").value);
   let attendance = Number(document.getElementById("attendance").value);

   // Validation

   if (
      empName === "" ||
      empId === "" ||
      department === "" ||
      designation === ""
   ) {

      alert("Please fill all employee details.");

      return;

   }

   let scores = [
```

```
    technical,
    communication,
    teamwork,
    productivity,
    attendance
];

// Validate Scores

for (let i = 0; i < scores.length; i++) {

    if (isNaN(scores[i]) || scores[i] < 0 || scores[i] > 100) {

        alert("Please enter valid scores between 0 and 100.");

        return;

    }

}

// Calculations

let total = scores.reduce((sum, score) => sum + score, 0);

let average = total / 5;

let highest = Math.max(...scores);

let lowest = Math.min(...scores);

// Performance Status

let status = "Satisfactory";

for (let i = 0; i < scores.length; i++) {

    if (scores[i] < PASS_SCORE) {

        status = "Needs Improvement";

        break;

    }

}

// Grade

let grade;
```

```
if (average >= 90) {
   grade = "A+";
}
else if (average >= 80) {
   grade = "A";
}
else if (average >= 70) {
   grade = "B+";
}
else if (average >= 60) {
   grade = "B";
}
else if (average >= 50) {
   grade = "C";
}
else {
   grade = "D";
}
// HR Remarks
let remarks;
switch (grade) {
   case "A+":
      remarks = "Outstanding Performer";
      break;
   case "A":
```

```
        remarks = "Excellent Performance";
        break;
    case "B+":
        remarks = "Very Good Performance";
        break;
    case "B":
        remarks = "Good Performance";
        break;
    case "C":
        remarks = "Average Performance";
        break;
    default:
        remarks = "Training Required";
}
// Promotion Eligibility
let promotion;
if (average >= 85) {
    promotion = "Eligible";
}
else {
    promotion = "Not Eligible";
}
// Bonus Eligibility
let bonus;
if (average >= 80) {
```

```
    bonus = "Eligible";

}

else {

    bonus = "Not Eligible";

}

// Object

const employee = {

    empName,
    empId,
    department,
    designation,
    total,
    average,
    highest,
    lowest,
    grade,
    status,
    promotion,
    bonus,
    remarks

};

// Object Destructuring

const {

    total: totalScore,
    average: avgScore,
    highest: highestScore,
    lowest: lowestScore,
    grade: finalGrade,
    status: finalStatus,
    promotion: promotionStatus,
    bonus: bonusStatus,
    remarks: hrRemarks

} = employee;

// Display Output

document.getElementById("total").innerHTML = totalScore + " / " + MAX_SCORE;
```

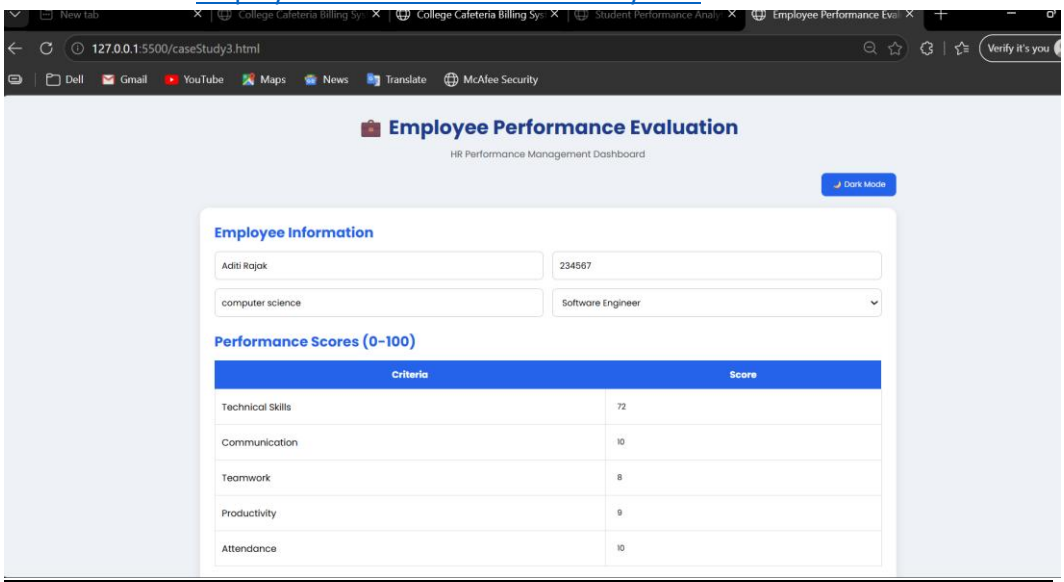
```
    document.getElementById("average").innerHTML = avgScore.toFixed(2) + "%";
    document.getElementById("grade").innerHTML = finalGrade;
    document.getElementById("status").innerHTML = finalStatus;
    document.getElementById("highest").innerHTML = highestScore;
    document.getElementById("lowest").innerHTML = lowestScore;
    document.getElementById("promotion").innerHTML = promotionStatus;
    document.getElementById("bonus").innerHTML = bonusStatus;
    document.getElementById("remarks").innerHTML = hrRemarks;
    // Progress Bar
    let progress = document.getElementById("progressBar");
    progress.style.width = avgScore + "%";
    progress.innerHTML = avgScore.toFixed(0) + "%";
    // Status Color
    if (finalStatus === "Satisfactory") {
        document.getElementById("status").style.color = "green";
    }
    else {
        document.getElementById("status").style.color = "red";
    }
    // Template Literal
    console.log(`Employee ${empName} secured ${avgScore.toFixed(2)}%
performance score.`);
    console.table(employee);
}
// Reset Form
```

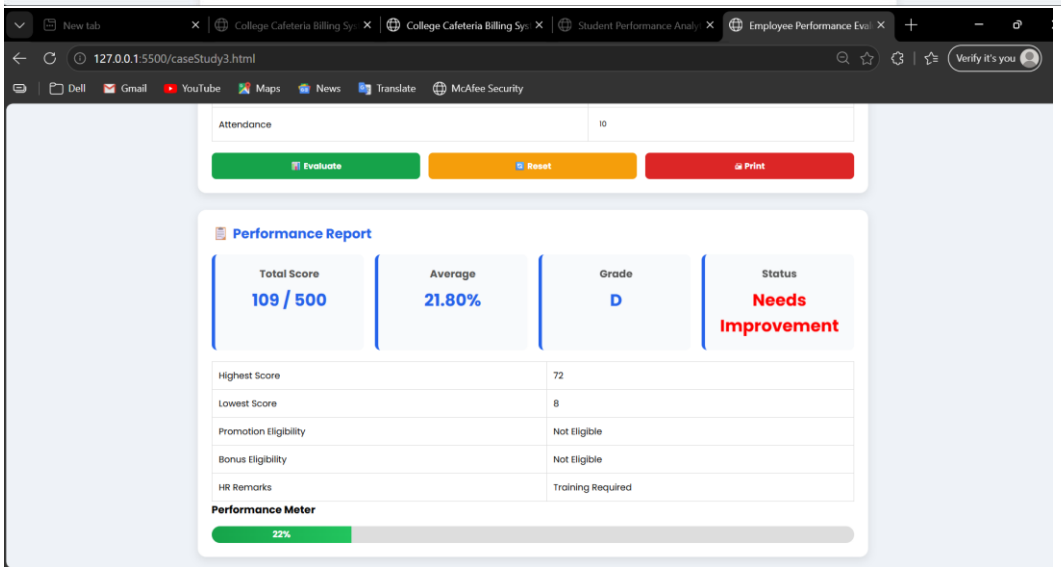
```
function resetForm() {
    document.getElementById("empName").value = "";
    document.getElementById("empId").value = "";
    document.getElementById("department").value = "";
    document.getElementById("designation").selectedIndex = 0;
    document.getElementById("technical").value = "";
    document.getElementById("communication").value = "";
    document.getElementById("teamwork").value = "";
    document.getElementById("productivity").value = "";
    document.getElementById("attendance").value = "";
    document.getElementById("total").innerHTML = "0";
    document.getElementById("average").innerHTML = "0";
    document.getElementById("grade").innerHTML = "-";
    document.getElementById("status").innerHTML = "-";
    document.getElementById("highest").innerHTML = "-";
    document.getElementById("lowest").innerHTML = "-";
    document.getElementById("promotion").innerHTML = "-";
    document.getElementById("bonus").innerHTML = "-";
    document.getElementById("remarks").innerHTML = "-";
    let progress = document.getElementById("progressBar");
    progress.style.width = "0%";
    progress.innerHTML = "0%";
}
// Print Report
function printReport() {
```

```
    window.print();
}
// Dark Mode
function toggleTheme() {
    document.body.classList.toggle("dark");
}
```

7. Output:

Employee Performance Evaluation System





8. Result/Conclusion:

This project successfully demonstrates the use of **JavaScript control structures** such as **if-else**, **switch**, **loops**, and **form validation** to automate the performance evaluation process. It reduces manual effort, improves accuracy, and provides a simple, interactive, and user-friendly solution. The project fulfills the objectives of the practical and can be enhanced further by adding a database and graphical reports.